

APPLIED RESEARCH

Textile Defect Detection Algorithm Based on the Improved YOLOv8

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ABSTRACT Automatic detection of textile defects is a crucial factor in improving textile quality. Fast and accurate detection of these defects is key to achieving automation in the textile industry. However, the detection of textile defects faces challenges such as small defect targets, low contrast between defects and the background, and significant variations in the aspect ratio of defects. To address these issues, this study proposes a new method for textile defect detection based on an improved version of You Only Look Once Version 8 (YOLOv8) called DA-YOLOv8s. Deep & Cross Network (DCNv2) is introduced into the Backbone Network to replace the C2F module, enhancing the extraction of network features; an self-attention mechanism, Polarized Self-Attention (PSA), is adopted to increase feature fusion capability and reduce feature loss in both channel and spatial dimensions; finally, a Small Object Detection Head (SOHead) is added to improve the feature extraction ability for small targets. Experimental results show that the improved YOLOv8 algorithm achieves has achieved mAP@0.5 and mAP of 44.6% and 48.6% respectively, which is an improvement of 4.2% and 3.8% over the original algorithm, and also outperforms the Optimal YOLOv9s model and the latest YOLOv11s model in these two metrics. The speed of textile defect detection has reached 257.38 frames per second (FPS) and the floating-point operation speed is 36.6 GFLOPS, ensuring the accuracy and speed of textile defect detection, with practical engineering application value.

INDEX TERMS Interest point detection, textile industry, quality management, YOLOv8, textile defect detection, polarized self-attention, deep & cross network.

I. INTRODUCTION

In the transformation and upgrading of the textile industry, the automated detection of defects and minor damages in textiles is an important field of industrial upgrading, as well as a significant factor in improving the quality of textiles. Traditional defect detection in textiles relies on manual visual inspection, that is, through the naked eye observation of inspectors, which is very limited in effectiveness and has a high rate of false positives and false negatives. According to statistics, manual inspection can detect about 40%-60% defects [1]. Therefore, how to detect textile defects

through automated means has become a research hotspot and difficulty in the textile industry.

With the development of visual technology, defect detection based on computer vision technology has become an important field in the research of automatic fabric defect detection [2]. The most commonly used computer vision methods include statistical feature methods, spectral analysis methods, model-based detection methods, and deep learning methods.

The principle of the statistical feature method is to study the pixel grayscale values in the texture of textile images, compare the differences in statistical features of different regions, and thus determine whether a region is a defect point area. Zhang et al. [3] proposed a multi-window grayscale ratio (MWGR) method for detecting defects in

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curtain cloth, which divides the image into several regions and analyzes the changes in grayscale ratio to determine the defects in the window of textiles. Zhu et al. [4] used the autocorrelation function to determine the pattern cycle of colored fabrics to determine the size of the detection window, then calculated the Gray Level Co-occurrence Matrix (GLCMs) to characterize the original image, and computed the Euclidean distance of GLCMs between the images to be detected and the defect-free template image to achieve defect detection. Guan et al. [5] first highlighted the defect areas using image enhancement techniques, then used the first-order derivative for edge detection while employing the Roberts operator to detect the edges of the defect areas to improve detection accuracy.

Spectral analysis methods treat images as two-dimensional signals with amplitude variations and perform frequency domain analysis through certain transformation algorithms, commonly including Fourier transform, Wavelet transform, and Gabor filter transform. Hu et al. [6] proposed an unsupervised method based on the combination of the discrete Fourier transform (DFT) and the discrete wavelet transform (DWT), which performs wavelet shrinkage denoising on the residual image after Fourier recovery, and applies inverse transformation to the approximation coefficients and processed wavelet coefficients separately to achieve defect information segmentation using simple thresholding. Li and Zhou [7] proposed a Defect Direction Projection Algorithm (DDPA) based on fabric defects characteristics, which filters the input image using Gabor filters, performs a Radon transform projection after using hard threshold segmentation, and selects the optimal Gabor filter channel, that is, the channel with the maximum defect value, to detect defects. Xiang et al. [8] proposed a defect detection algorithm based on Fourier convolution, which generates image pairs for training using random masking in the training phase and incorporates a Fourier convolution layer into the autoencoder to achieve automatic detection of dyed fabric defects.

The defect detection algorithms based on traditional computer vision have high computational requirements and need to be improved in terms of detection speed and accuracy.

Deep learning is a new framework in computer vision research, which has been widely applied in the field of defect detection with the rapid development of big data and artificial intelligence technologies [9], [10]. Deep learning can automatically extract features, optimize and iterate parameters, thus achieving the function of detecting defects in textile images. Mei et al. [11] designed a Multi-Scale Convolutional Denoising Autoencoder Network (MSCDAE) that achieved unsupervised detection of textile defects. The algorithm trained the Convolutional AutoEncoder (CAE) with positive samples, enabling it to extract fabric features and reconstruct fabric images. Detection is realized by identifying defects based on the difference in features between defective images and normal fabric images. Jing et al. [12] proposed an automatic detection method for fabric defects

based on convolutional neural networks. This method decomposes textile images into multiple local patches and labels them, then transmits them to a pre-trained deep CNN for learning, and uses the trained model to detect each patch, thereby obtaining the category and position of each defect. Ma et al. [13] used an improved parameter VGG16 model to train a classifier for detecting and recognizing defects in denim fabrics and constructed a defect detection algorithm based on a cascading architecture by merging the two models. Deep learning methods lack in speed and accuracy.

In recent years, with the further development of computer vision technology, research on object detection algorithms has largely focused on those based on candidate regions and regression-based deep convolutional neural networks. The Faster R-CNN algorithm is a representative of the candidate region-based object detection algorithms, demonstrating excellent performance in the field of object detection [14], [15], [16], [17]. Wei et al. [18] proposed a Faster R-CNN model based on an improved VGG structure, which adapts to the characteristics of fabric defect images by reducing the number of anchor points in the Faster R-CNN. The VGG16 was modified to include 13 convolutional layers (with the activation function being ReLu) and four pooling layers to extract feature maps. Additionally, the Region Proposal Network (RPN) and the ROI pooling layer were improved to enhance the model. An et al. [19] improved the Faster R-CNN network for textile defect detection by using deep residual networks instead of traditional VGG-16 for feature extraction, and incorporating methods such as adding feature pyramid modules and increasing the number of anchor boxes. Chen et al. [20] designed a Genetic Algorithm Gabor Faster R-CNN (Faster GG R-CNN) model, which embeds Gabor kernels into Faster R-CNN and employs a two-stage training method based on Genetic Algorithm (GA) and backpropagation for textile defect detection. The training of the Faster R-CNN algorithm is a two-stage object detection algorithm, with the first stage completing region box proposals and the second stage conducting object recognition within the region boxes, which affects the speed and accuracy of object detection.

Regression-based object detection algorithms directly regress the bounding box coordinates and object categories at multiple positions in the input image, addressing the coexistence of accuracy and speed issues; YOLO is a typical representative of such algorithms. The diversity of YOLO's applications in industrial defect detection also verifies the effectiveness of the algorithm [21], [22], [23], [24]. Yue et al. [25] proposed an improved YOLOv4 textile defect detection algorithm, which, based on the expansion of the dataset using combined data augmentation methods, improved the head prediction layer and integrated the Convolutional Block Attention Module (CBAM) to achieve accurate classification and localization of tiny target defects. Jin and Di [26] also made improvements to the YOLOv5 network, introducing spatial and channel attention models into the backbone

network and designing a multi-task learning strategy with two detection heads for detecting common defects and identifying specific defects to improve the accuracy of defect recognition. These methods have weaker detection capabilities for irregularly sized defects, and the accuracy can also be further enhanced.

Section I in our paper introduces the background of textile defect detection and the development of detection methods, summarizes the characteristics of these methods, and proposes improvements and innovations to the YOLOv8s benchmark model. Section II describes the principles of convolutional neural networks, the feasibility of incorporating attention mechanisms into neural networks, and introduces the basic network structure of YOLOv8. Section III specifically proposes the construction of DA-YOLOv8s model, including the introduction of DCNv2 to enhance the feature extraction capability of the backbone network, the introduction of PSA to improve the feature fusion capability of the network neck, and the addition of SOHead detection head to enhance the detection capability of small targets. Section IV introduces the dataset and evaluation metrics, conducts comparative experiments and ablation study, and analyzes the experimental results. Section V summarizes the work achievements of this paper and looks forward to future research. The structure of the paper content is shown in Fig 1.

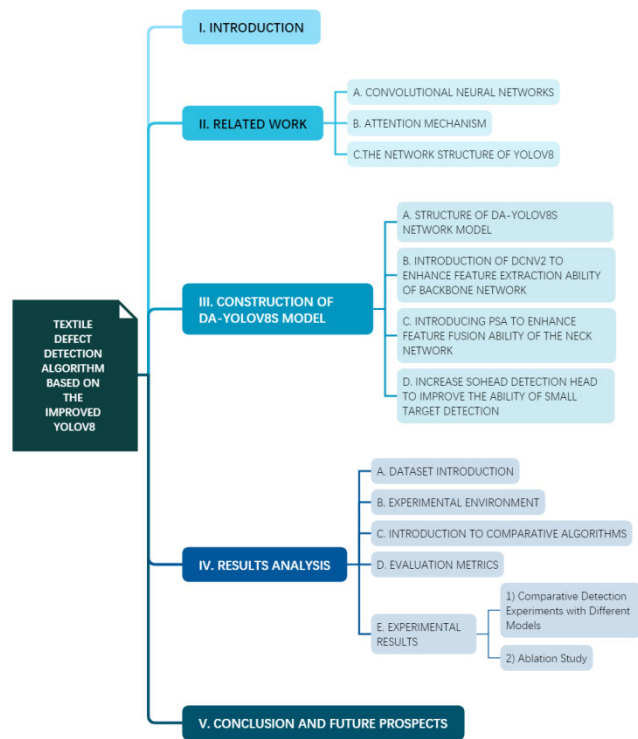


FIGURE 1. Paper content structure diagram.

YOLOv8 is an open-source object detection model released in January 2023. Our research focuses on improving the accuracy of textile defect detection using deep neural network technology, and we propose a method called DA-

YOLOv8s. We take the currently popular object detection algorithm YOLOv8 as the baseline model and improve the traditional convolutional neural network in the C2F module of YOLOv8's backbone network using the Deep and Cross Network. Additionally, we introduce the PSA attention mechanism to enhance YOLOv8's neck network. At the same time, we incorporate a small object detection head in YOLOv8's head module to improve feature extraction and fusion capabilities, thereby enhancing the accuracy of textile defect detection. The main contributions of our paper are as follows:

- (1) We incorporate DCNv2 (Deep & Cross Network) [27] into the Backbone to replace the C2f module, enhancing the model's feature extraction capability.
- (2) We introduce a new feature fusion algorithm, PSA (Polarized Self-Attention) [28], which uses an extreme self-attention mechanism to reduce feature loss in channel and spatial dimensions.
- (3) We add a detection head for small objects SOHead to prevent the loss of small object features, thereby improving detection performance.

Experiments show that using YOLOv8s as the base network model achieves textile defect detection and recognition, with improved DA-YOLOv8s increasing mAP@0.5 and mAP@0.5,0.3,0.1 by 4.2% and 3.8%, reaching 0.446% and 48.6%, but there is still room for improvement in terms of model accuracy.

II. RELATED WORK

In order to enhance the feature extraction capabilities of computer images and improve detection accuracy, extensive research has been conducted. In this section, we introduce the relevant research work and, in combination with various studies, propose improved methods. The methods and their performance are shown in the Table 1.

TABLE 1. Related studies and improvement methods.

Research Focus	Improved Method	mAP@0.5	mAP
CNN	DCNv2	42.7%(+2.3%)	47.4%(+2.6%)
Attention Mechanism	PSA	41.5%(+1.1%)	46.1%(+1.3%)
YOLOv8	DCNv2+PSA+SOHead	44.7%(+4.2%)	48.6%(+3.8%)

A. CONVOLUTIONAL NEURAL NETWORKS

Convolutional Neural Networks (CNNs), as one of the most significant algorithms in computer vision, were successfully applied in the LeNet network structure in 1998 for recognizing handwritten digits [29], demonstrating the powerful capabilities of CNNs in image processing. Since then, CNNs have been widely used in various fields such as object detection and classification algorithms. A CNN is a feedforward neural network that includes convolutional computations and deep architectures, with its core concept being the extraction of features through operations such as convolution and pooling. The components of a CNN include:

Input Layer: This layer standardizes the input image data and performs data augmentation operations to ensure that the input data meets the requirements of the convolutional layer.

Convolutional Layer: This layer performs convolutional operations and feature extraction on the input image. Each element of the convolutional kernel corresponds to a weight coefficient and a bias vector. When operating, the convolutional kernel systematically sweeps across the input features, performing matrix element multiplication and summation within the receptive field, and adding the bias term.

Activation Function: Its primary role is to apply a non-linear mapping to the output of the convolutional layer, enabling the network to have better learning capabilities. Commonly used activation functions include Sigmoid, Tanh, ReLU, etc.

Pooling Layer: After feature extraction by the convolutional layer, the output feature maps are passed to the pooling layer for feature selection and information filtering. The pooling layer contains pre-defined pooling functions, which replace the result of a single point in the feature map with a statistical measure of its neighboring area. Common pooling functions include average pooling and max pooling strategies.

Fully Connected Layer: While the convolutional neural network is capable of feature extraction from input data, the role of the fully connected layer is to perform non-linear combinations of the features extracted by the convolutional and pooling layers to produce the output.

Output Layer: For image classification problems, the output layer uses a logistic function or a normalized exponential function (softmax function) to output classification labels. In object detection problems, the output layer can be designed to output the center coordinates, size, and classification of the object.

Traditional CNNs extract features in the form of linear models, which have limited extraction capabilities. In contrast, the Cross Network can achieve multi-layer feature interactions, with each layer producing higher-order interactions based on existing ones, and retaining interactions from previous layers. The cross network can be trained jointly with a deep neural network [30]. Here, we introduce the DCNv2 model to improve the C2f in YOLOv8, which includes the classic CNN, as discussed in Section III, Part B.

B. ATTENTION MECHANISM

The Attention Mechanism is an approach that emulates the human visual and cognitive systems. By incorporating attention mechanisms into neural networks, the networks can automatically learn to selectively focus on important information within the input data, thereby enhancing the model's performance and generalization capabilities. The most typical attention mechanisms in the field of computer vision include channel attention, spatial attention, and self-attention mechanisms. Channel attention adaptively recal-

brates the weight of each channel, which can be considered an object selection process, thus determining what to pay attention to [31]. Spatial attention can be regarded as an adaptive spatial region selection mechanism: where to pay attention. It emphasizes or suppresses information at these locations by assigning different weights to each pixel or pixel block. Self-attention mechanisms, on the other hand, automatically learn the associations between different positions, thereby capturing richer contextual information [32].

Attention mechanisms have achieved significant progress in computer vision applications and are often used in conjunction with classic object detection models. Therefore, based on the characteristics of textile defects, we introduce the Polarized Self-Attention(PSA) mechanism, which combines channel, spatial, and self-attention mechanisms, to improve the network structure of YOLOv8 and enhance algorithm performance. The specific principles are elaborated in Section III, Part C.

C. THE NETWORK STRUCTURE OF YOLOv8

The network structure of YOLOv8 can be divided into four parts: the input end, the backbone network, the neck network, and the head module, as shown in Fig 2. The input end mainly includes Mosaic data augmentation, adaptive anchor box calculation, and adaptive grayscale padding. The Backbone network contains structures such as Conv, C2f, and SPPF, among which the C2f module is the primary module for learning residual features, with branch connections across layers that enrich the gradient flow of the model and form a neural network module with stronger feature representation capabilities. The Neck network adopts the PAN (Path Aggregation Network) structure, which enhances the network's ability to fuse features of objects at different scaling scales. The Head module is the output end, which decouples the classification and regression processes; the loss calculation process mainly includes positive and negative sample assignment strategies and loss computation, with the assignment strategy using the dynamic assignment method of Task Aligned Assigner [33], which selects positive samples based on the weighted results of classification and regression scores. The loss calculation for the classification branch adopts the BCE Loss method, while the regression branch uses the distribution focal loss [34] and the CIoU (complete intersection over union) loss function algorithm. The network structure is shown in the Fig 2.

III. CONSTRUCTION OF DA-YOLOv8s MODEL

Our study adopts YOLOv8s as the baseline model, with the aim of enhancing the precision of textile defect detection while also meeting the speed requirements for industrial-grade applications. To achieve these objectives, we propose an algorithm called DA-YOLOv8s. This section will provide a detailed introduction to the model architecture of DA-YOLOv8s and its innovative points of improvement.

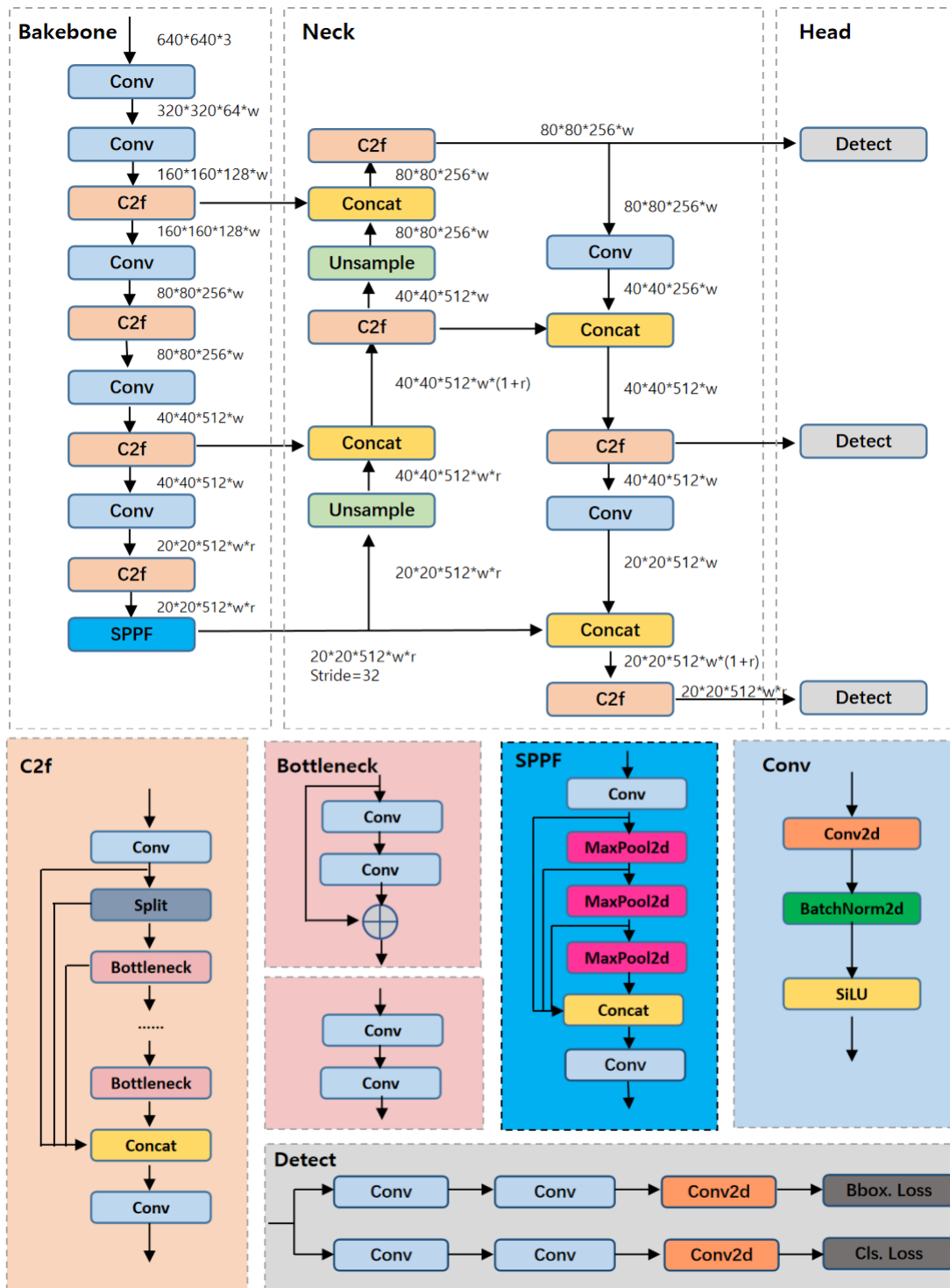


FIGURE 2. YOLOv8 network structure diagram. Here w represents the width of the convolutional kernel, r represents the scale factor.

A. STRUCTURE OF DA-YOLOv8s NETWORK MODEL

To address issues such as small defect targets, low contrast between defects and background, and variable aspect ratios of defects in textile defect detection, this study proposes a new textile defect detection method, DA-YOLOv8, based

on an improved YOLOv8s. By incorporating DCNv2 into the Backbone network to replace the C2f module, adopting the Polarized Self-Attention (PSA) mechanism to improve the Neck network, and introducing the SOHead detection head for small objects, the improved DA-YOLOv8s

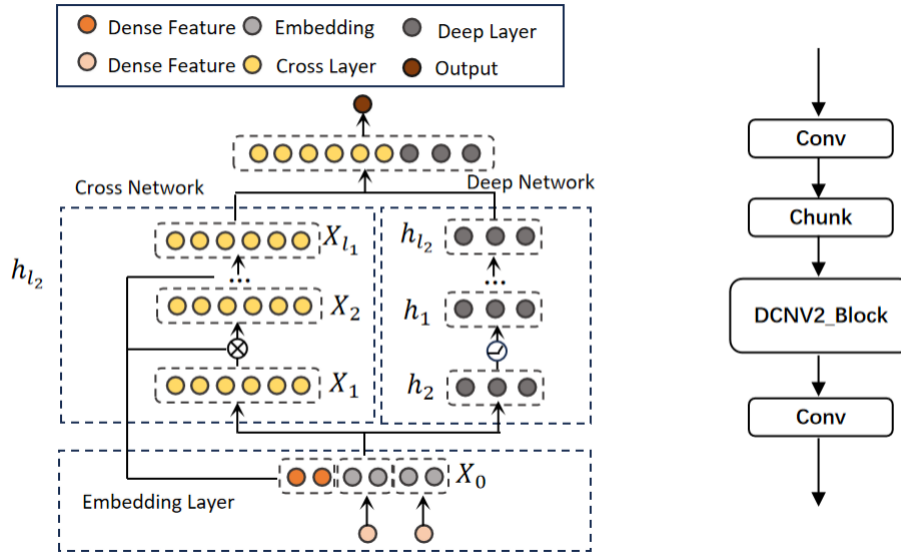


FIGURE 4. The network structure of DCNv2. The left figure details the specific algorithm of DCNv2_Block.

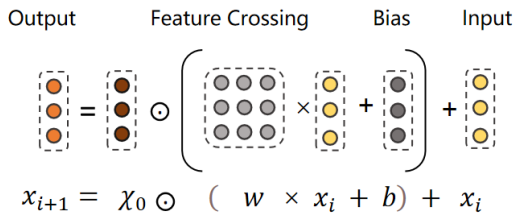


FIGURE 5. Single-layer operation rule for cross networks.

Deep and cross combination: The combination of cross networks and deep networks results in two structures, namely the stacked structure and the parallel structure. The stacked structure feeds the output of the cross network into the deep network as its input. The parallel structure, however, processes the two networks in parallel and finally combines their outputs with a single output layer. In practice, which architecture performs better depends on the data.

The formula for the predictive function is as shown in (2):

$$\hat{y}_i = \sigma \left(W_{\text{logit}}^T x_{\text{final}} \right) \quad (2)$$

where W_{logit} is the weight vector for the logit, $\sigma(x) = 1 / (1 + \exp(-x))$. For the final loss, the logarithmic loss function Log Loss is used, as shown in (3).

$$\begin{aligned} \text{loss} &= -\frac{1}{N} \sum_{i=1}^N y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) + \lambda \sum_l \|W_l\|_2^2 \end{aligned} \quad (3)$$

where $\hat{y}_i$ is the prediction; y_i is the true label; N is the total number of inputs; λ is the L_2 regularization parameter.

C. INTRODUCING PSA TO ENHANCE FEATURE FUSION ABILITY OF THE NECK NETWORK

The Neck network in YOLOv8, through a series of convolutional layers and upsampling layers, fuses low-level feature maps with high-level feature maps, thereby enhancing the accuracy of object detection. However, to more effectively capture the details of objects and the feature information within small target regions, with a focus on feature extraction at the channel and specific spatial levels. In response to this issue, Polarized Self-Attention (PSA) is introduced to enhance the representation fusion capability of the entire network.

Polarized self-attention mechanisms are used to address pixel-level regression tasks, maintaining relatively high resolution in both channel and spatial dimensions (retaining $C/2$ dimensions in the channel and $[H, W]$ dimensions in the spatial dimension), which can reduce information loss caused by dimensionality reduction; it also synthesizes nonlinear functions that directly correspond to the typical fine-grained regression output distribution, making the fitted output more refined and closer to the actual output. This structure includes self-attention mechanisms in both channel and spatial dimensions and fuses the results of these two dimensions to obtain the polarized self-attention output. There are two structures of the polarized attention mechanism, namely the parallel structure and the sequential structure, as shown in the left figure in Fig 6 and Fig 7 respectively. In this project, the parallel structure is integrated into the YOLOv8 network. Within the Neck network, the PSA module receives the output from the C2F module and

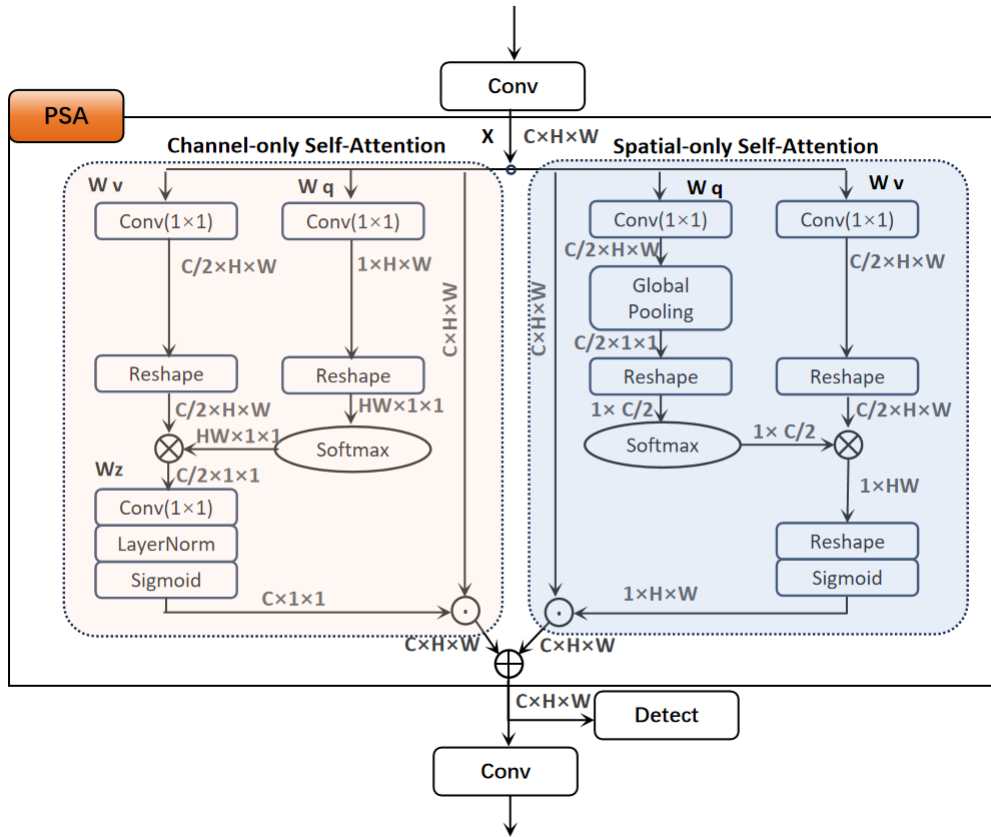


FIGURE 6. Polarized self-attention module with a parallel structure in YOLOv8.

processes it in parallel using channel-wise self-attention and spatial self-attention, performing convolutions, reshaping, and applying the Sigmoid function, among other operations. The results of these computations are then combined and output to the detection head and Conv modules. The enhanced structure of the neck network in YOLOv8 is illustrated in Fig 6. The PSA computation process is as follows:

Channel Dimension Self-Attention $A^{ch}(X) \in \mathbb{R}^{C \times 1 \times 1}$: First, the input features X are transformed into W_q and W_v using the convolution of 1×1 , where the channels of W_q are fully compressed, while the channel dimension of W_v remains at a relatively high level ($C/2$). Because the channel dimension of W_q is compressed, information enhancement is required through HDR, so the information of W_q is enhanced using Softmax. Then, W_q and W_v are subjected to matrix multiplication, followed by a 1×1 convolution and LN to increase the dimension on the channels to C . Finally, the Sigmoid function is used to keep all parameters between 0-1. The operation is as shown in (4):

$$A^{ch}(X) = F_{SG} \left[W_{z|\theta_1} \left(\sigma_1(W_v(X)) \times F_{SM}(\sigma_2(W_q(X))) \right) \right] \quad (4)$$

where W_q , W_v and W_z are the convolutional layers 1×1 , σ_1 and σ_2 are two tensor reshaping operators, $F_{SM}(\cdot)$ is the SoftMax operator, “ $\times$ ” is the matrix dot product operation as (5). The number of internal channels between W_v | W_q and W_z is $C/2$, and the output of channel self-attention is $Z^{ch} = A^{ch}(X) \odot^{ch} X \in \mathbb{R}^{C \times H \times W}$, where ch is the channel multiplication operator.

$$F_{SM}(X) = \sum_{j=1}^{N_p} \frac{e^{x_j}}{\sum_{m=1}^{N_p} e^{x_m}} x_j \quad (5)$$

Spatial Dimension Self-Attention $A^{sp}(X) \in \mathbb{R}^{C \times H \times W}$: First, the input features were transformed into W_q and W_v using convolution of 1×1 . For W_q features, Global Pooling was applied to compress the spatial dimension, transforming it into the size of 1×1 ; while the spatial dimension of W_v features was maintained at a relatively large level ($H \times W$). Since the spatial dimension of W_q was compressed, Softmax was used to enhance the information of W_q . Finally, matrix multiplication was performed on W_q and W_v , followed by reshape and Sigmoid to ensure all parameters remained

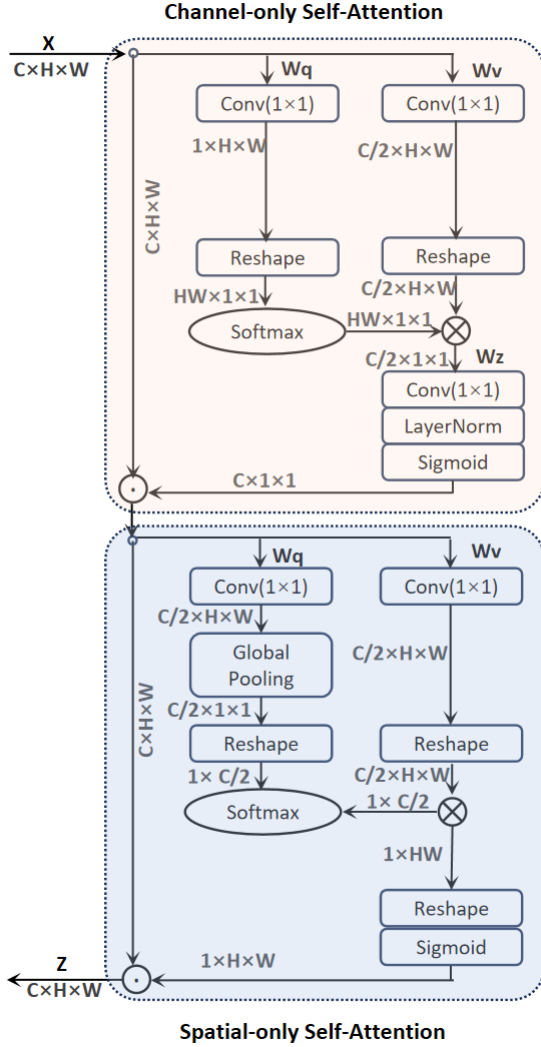


FIGURE 7. Polarized self-attention module in serial structure.

between 0-1. The operation is as shown in (6).

$$A^{sp}(X) = F_{SG}[\sigma_3(F_{SM}(\sigma_1(F_{GP}(W_q(X)))) \times \sigma_2(W_v(X)))] \quad (6)$$

In the formula, W_q and W_v are standard 1×1 convolutional layers, σ_1, σ_2 and σ_3 are three tensor reshaping operators, $F_{SM}(\cdot)$ is the SoftMax operator. $F_{GP}(\cdot)$ is a global pooling operator as (7). “ $\times$ ” denotes the dot product operation of matrices. The output of spatial self-attention is $Z_{sp} = A_{sp}(X) \odot^{sp} X \in \mathbb{R}^{C \times H \times W}$, where sp is the spatial multiplication operator.

$$F_{GP}(X) = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W X(:, i, j) \quad (7)$$

Combination of Channel and Spatial Self-Attention: The parallel combination of channel and spatial self-attention

forms a PSA parallel structure, as shown in (8).

$$\begin{aligned} PSA_P(X) &= Z^{ch} + Z^{sp} \\ &= A^{ch}(X) \odot^{ch} X + A^{sp}(X) \odot^{sp} X \end{aligned} \quad (8)$$

The sequential combination of channel and spatial self-attention forms a PSA series structure, as shown in (9).

$$\begin{aligned} PSA_P(X) &= Z^{sp}(Z^{ch}) \\ &= A^{sp}(A^{ch}(X) \odot^{ch} X) \odot^{sp} A^{ch}(X) \odot^{ch} X \end{aligned} \quad (9)$$

D. INCREASE SOHead DETECTION HEAD TO IMPROVE THE ABILITY OF SMALL TARGET DETECTION

The classic YOLOv8 has three detection heads, which can detect targets at multiple scales, with detection sizes of:

The corresponding detection feature map size for P3/8 is 80×80 , used for detecting targets larger than 8×8 .

The corresponding detection feature map size for P4/16 is 40×40 , used for detecting targets larger than 16×16 .

The corresponding detection feature map size for P5/32 is 20×20 , used for detecting targets larger than 32×32 .

However, there may be missed detections or poor detection ability for tiny targets, hence adding a detection head for small objects, maintaining the extraction of small target features through downsampling.

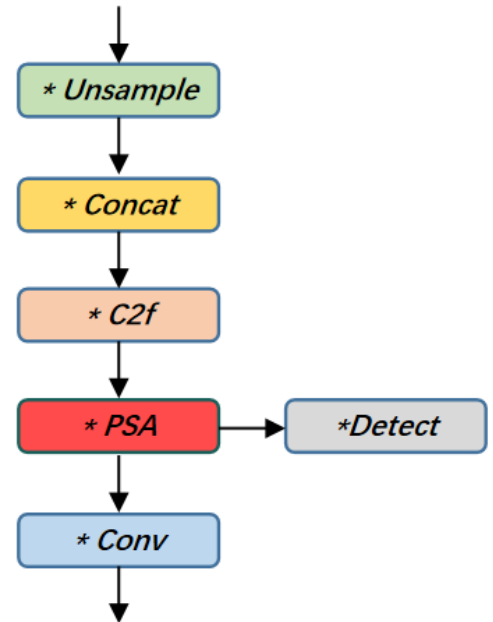


FIGURE 8. Modules added for small target detection.

The newly added detection feature map for 160×160 is used for detecting targets larger than 4×4 . Upon the addition of this detection head, the Neck network necessitates the augmentation of a corresponding detection module. In this context, we incorporate the Unsample, Concat, C2f, and PSA modules in sequential order to perform computations.

The outcomes of these operations are then fed into a Conv module for continued feature extraction, and concurrently into a Detect module to finalize the object detection process. The network architecture is depicted in Fig 8.

IV. RESULTS ANALYSIS

Our study used the Tianchi 2019 Guangdong Industrial Intelligent Manufacturing Innovation Competition dataset to conduct comparative experiments with DA-YOLOv8s and ten other models, including a baseline model. Ablation experiments were also performed. The results indicated that the DA-YOLOv8s model showed a significant improvement in mAP and achieved a balance between accuracy and speed.

A. DATASET INTRODUCTION

The dataset is from the Tianchi 2019 Guangdong Industrial Intelligent Manufacturing Innovation Competition. It is a textile defect dataset created from images collected at a textile workshop. The dataset contains a total of 4774 images, with image sizes of 2446 px * 1000 px. There are 3819 images in the training set and 955 images in the validation set. It covers 20 important categories of fabric defects in the textile industry, with a total of 6126 defect instances, and provides detailed annotations. Each image contains one or more defect instances, with some typical defects shown in Fig 9.

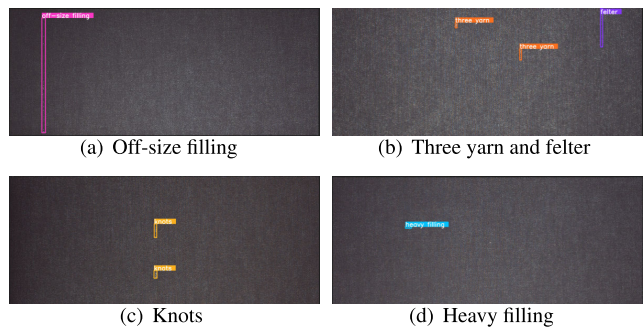


FIGURE 9. Visual comparisons of original models.

The dataset has an imbalanced distribution of categories, with the ‘knots’ defect category exceeding 1200 instances, while the ‘centipede’ category has only 79 instances. The statistics of each defect category are shown in Fig 10; we have normalized the standard size of defect points, and the distribution of the length and width of defect points is shown in Fig 11. It can be observed from the figure that there is a significant size variation among defects in the dataset, and the detection target is relatively small compared to the original image. The proportion of defect point widths to the width of the original image is within 0.1, accounting for 86.0%.

B. EXPERIMENTAL ENVIRONMENT

The experimental environment for this paper is set up on a server with the following hardware configuration as Table 2: Jensen RTX 4090 24GB graphics card, 64 Gb memory.

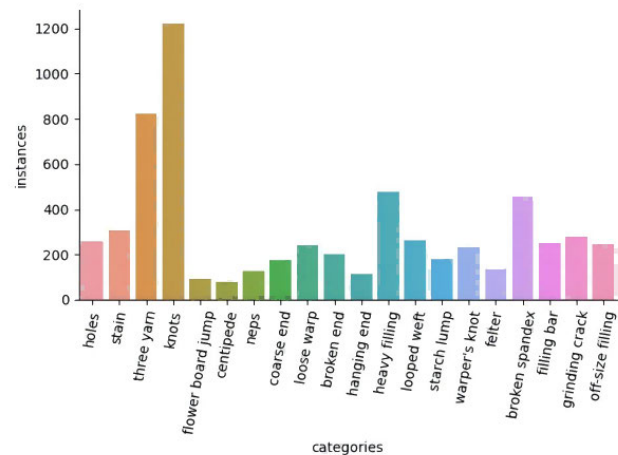


FIGURE 10. Statistics of defect points by category.

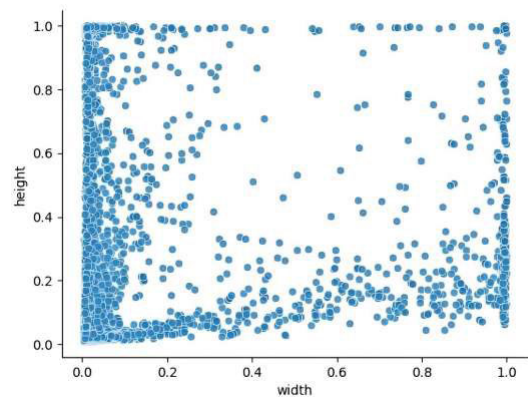


FIGURE 11. Scatter plot of defect sizes.

The server system is configured with Ubuntu 20.04 version, CUDA Toolkit 12.0 version, and the deep learning framework platform is Pytorch 2.0.1 version.

TABLE 2. Experimental environment configuration.

Software/Hardware	Version Information
Operating System	Ubuntu 20.04
Programming Language	Python
CUDA Version	CUDA12.0
Deep Learning Framework	Pytorch 2.0.1
GPU Model	RTX 4090 24GB

In the experiment, parameters are set according to the default hyperparameters of the model, batch size (batch size) is 16, the number of training periods (epochs) is 100, and the initial learning rate (learning rate) is 0.0001, as shown in Table 3.

C. INTRODUCTION TO COMPARATIVE ALGORITHMS

To verify the effectiveness of the improved model, we compared it with the classical object detection algorithms Faster R-CNN [35], Cascade R-CNN [36], YOLOv3 [37],

TABLE 3. Parameter configuration.

Hyperparameter	Value
Batch Size	16
Epochs	100
Learning Rate	0.0001

YOLOv5, YOLOX [38], YOLOv6 [39], YOLOv9 [40], YOLOv10 [41] and YOLOv11 [42], under the experimental environment consistent with Table 3, using the default parameters of the model.

Faster R-CNN Algorithm Model: Faster R-CNN is a classical object detection algorithm proposed by Ross Girshick in 2016. Faster R-CNN uses VGG16 as the network's backbone, first extracting features with a set of basic conv+relu+pooling layers, then passing through the RPN (Region Proposal Networks) to generate candidate boxes based on the Anchor mechanism. Finally, feature extraction, candidate box selection, bounding box regression, and classification are integrated into one network, thus providing higher detection accuracy and efficiency.

The Cascade R-CNN Algorithm Model: This represents an enhancement over the Faster R-CNN, proposing a cascaded R-CNN detection and inference architecture composed of a sequence of detectors trained with increasing IoU thresholds. This cascaded sampling progressively enhances the quality of detection.

The YOLOv3 Algorithm Model: Introduced in 2018, the YOLO detection model utilizes a Feature Pyramid Network (FPN) for feature fusion, incorporating residual connection modules to enable multi-scale training. It outputs feature maps at three scales, and in this context, we adopt the YOLOv3-tiny as the baseline model.

YOLOv5 algorithm model: YOLOv5 is an object detection model released by Ultralytics in June 2020, which excels in inference speed. This paper adopts the YOLOv5s model version, which has a smaller number of parameters and is suitable for lightweight resource devices or scenarios requiring fast inference. The YOLOv5 network structure consists of an input end (Input), a Backbone network, a Neck network, and a detection end (Head). The input end uses data augmentation techniques and adaptive anchor calculation methods to enrich the dataset and reduce the occupation of GPU resources; the Backbone network utilizes the Focus and CSPDarknet53 structures to optimize the classifier, enhancing the diversity and robustness of features; the Neck network adopts the SPP module and FPN+PAN structure, which strengthens semantic and positional dual information, fuses feature information extracted by the Backbone network, and further improves the model's performance and accuracy.

YOLOX algorithm model: YOLOX is an object detection algorithm proposed by Megvii Technology in 2021. It is improved based on YOLOv3-SPP, adopting the design idea of decoupled heads, with classification prediction and position prediction handled by two separate networks. This approach not only reduces information redundancy but

also improves detection accuracy. YOLOX introduces an anchor-free design concept, achieving anchor-free detection by directly predicting the center point coordinates and width and height information of objects, which not only enhances detection flexibility but also effectively avoids performance bottlenecks caused by anchor boxes. Moreover, YOLOX employs an advanced label assignment strategy, taking into account factors such as the size, position, and shape of objects, making label assignment more rational and accurate.

YOLOv6 algorithm model: YOLOv6 is an object detection framework developed by Meituan's Visual Intelligence Department and released in 2022. YOLOv6 has made numerous improvements to network structures such as Backbone, Neck, and Head. In this paper, YOLOv6 also uses the YOLOv6s version as the comparison algorithm. The Backbone network in YOLOv6s is inspired by the RepVGG [43] Style structure, composed of RepBlock in the training phase, and in the inference phase, each RepBlock is converted into a 3*3 convolutional layer stack (represented as RepConv) through the ReLU activation function, which can reduce inference latency while enhancing representation capabilities. The Neck network layer replaces the CSPBlock used in YOLOv5 with RepBlock, i.e., Rep-PAN, and correspondingly adjusts width and depth; the Head network adopts a hybrid-channel strategy to construct a more efficient decoupled head, which further reduces computational cost.

YOLOv9 introduced the concept of Programmable Gradient Information (PGI) to address the diverse changes required by deep networks to meet various objectives. Furthermore, YOLOv9 has developed a new lightweight network architecture, the Generalized Efficient Layer Aggregation Network (GELAN), which employs gradient path planning to significantly improve detection performance.

YOLOv10 has introduced a consistent dual assignment strategy with dual label assignments and a consistent matching metric to address the issue of redundant predictions in post-processing without the need for Non-Maximum Suppression (NMS). It has proposed a lightweight classification head, spatial-channel decoupled downsampling, and rank-guided block design to reduce explicit computational redundancy and achieve a more efficient architecture.

The YOLOv11 Algorithm Model: This is the latest iteration of the YOLO model, which builds upon YOLOv8 by refining the network architecture. It replaces the C2f module with the C3K2 module and adds an attention mechanism C2PSA module following the SPPF layer. Additionally, it improves the structure of the detection head. We have also introduced this model as a comparative model in our experimental evaluation.

D. EVALUATION METRICS

To validate the effectiveness and execution time of the model, this paper uses GFLOPS (Giga Floating-point Operations Per Second, representing 1 billion floating-point operations per second) as a measure of the execution time of the network

model; the mean average precision (mAP) metric is used to evaluate the accuracy of the model, with its calculation formula as shown in (10).

$$P_{mA} = \sum P_A / N \quad (10)$$

In the formula, mAP, N represents the total number of categories, and P_A is the area enclosed by the curve formed by recall on the horizontal axis and precision on the vertical axis. We use mAP@0.5 and mAP as evaluation metrics, where mAP@0.5 is the mean Average Precision at an Intersection over Union (IOU) threshold of 0.5, and mAP is the mean Average Precision at IOU thresholds of 0.5, 0.3, and 0.1.

Simultaneously, we employ FPS (Frames Per Second) to evaluate the detection speed of the model. FPS denotes the number of image frames that can be processed and outputted within a second. The calculation method is illustrated in (11). In this equation, t_1 represents the image preprocessing time, t_2 signifies the image inference time, and t_3 indicates the post-processing time.

$$FPS = \frac{1000 \text{ ms}}{t_1 + t_2 + t_3}; \quad (11)$$

The evaluation metrics are shown in the Table 4:

TABLE 4. Evaluation metrics functions and objectives.

Metric	Function	Objective
mAp	Evaluates the average precision of the model	To achieve a significant improvement
GFLOPS	Evaluates the computational complexity of the model	For reference, based on FPS
FPS	Evaluates the detection speed of the model	≥ 60 FPS

E. EXPERIMENTAL RESULTS

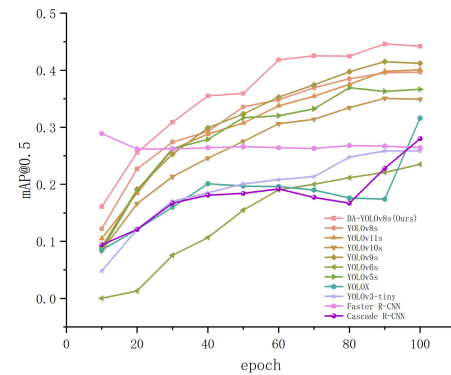
1) COMPARATIVE DETECTION EXPERIMENTS WITH DIFFERENT MODELS

To fully evaluate the detection algorithm of the improved YOLOv8s model in this paper, ten algorithms were selected for experimental comparison, including the Faster R-CNN, Cascade R-CNN, and unimproved YOLOV3-tiny, YOLOv5s, YOLOv6s, YOLOX, YOLOv8s, YOLOv9s, YOLOv10s, and YOLOv11s. Meanwhile, a comparison was made with the typical two-stage object detection algorithm, Faster-RCNN. The mean Average Precision (mAP) after 100 iterations was used as the evaluation criterion for different detection algorithms, which can scientifically and reasonably assess the object detection capability and computational speed of various detection algorithms. The results of the experimental comparison are shown in Table 5. The mean precision after each training iteration is shown in Fig 12. The results indicate that, considering the balance between detection accuracy and speed, the YOLOv8s, YOLOv9s, and YOLOv11s models significantly outperform other models. Using the improved DA-YOLOv8s model, which is superior to all the models it is compared with, there is a 4.2% increase in mAP@0.5 and

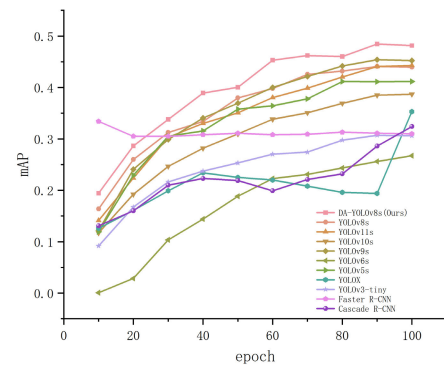
a 3.8% improvement in average mAP relative to the baseline YOLOv8s model, although the detection speed has decreased by 52.2%, and the GFLOPS has risen by 12.1. Given YOLO's excellent performance in detection speed, the DA-YOLOv8s still achieves a detection speed of 257.38 FPS, which meets the requirements for industrial applications.

TABLE 5. Comparison of detections for different models.

Model Name	mAP@0.5	mAP	FPS	GFLOPS
Faster R-CNN	0.296	0.339	85.76	157.7
Cascade R-CNN	0.255	0.295	90.82	187.0
YOLOv3-tiny	0.256	0.313	723.01	18.9
YOLOX	0.316	0.353	190.4	36.9
YOLOv5s	0.375	0.421	590.88	23.9
YOLOv6s	0.238	0.271	546.81	44.1
YOLOv9s	0.412	0.452	483.15	26.8
YOLOv10s	0.349	0.387	535.37	24.5
YOLOv11s	0.401	0.445	518.29	21.3
YOLOv8s	0.404	0.448	546.83	28.5
DA-YOLOv8s(Ours)	0.446	0.486	257.38	36.6



(a) Comparison of mAP@0.5 across different epochs



(b) Comparison of average precision mAP across epochs

FIGURE 12. Comparison chart of training accuracy across epochs.

The box_loss, cls_loss (Classification Loss), and dfl_loss (Distributional Feature Loss) of the DA-YOLOv8s model gradually decreased during the training process, as shown in Fig 13, indicating that the model's performance in identifying the location, size, and category of textile defects is improving. The consistency between training loss and validation loss, both showing a downward trend, suggests that the DA-YOLOv8s model has good generalization capabilities.

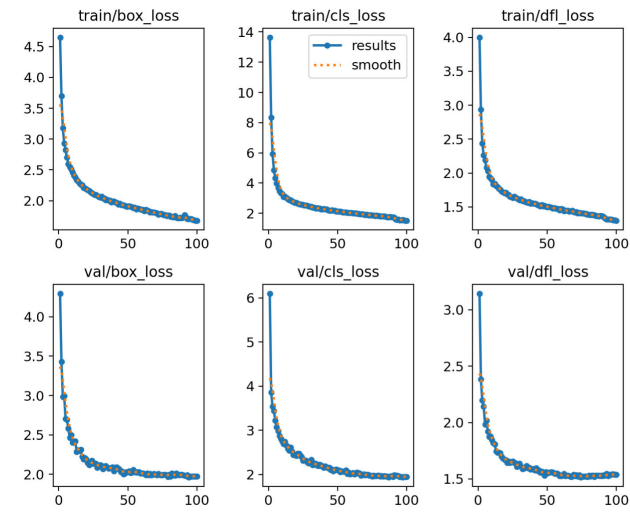


FIGURE 13. Box_loss, Cls_loss, and Dfl_loss metrics.

2) ABLATION STUDY

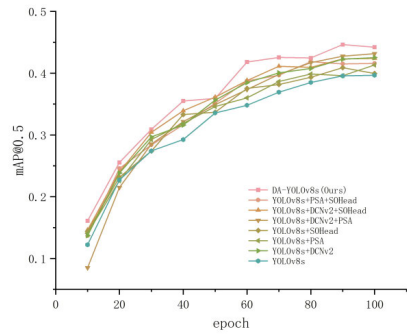
In the Backbone, the DCNv2 module is used to replace all original C2f modules of YOLOv8 for ablation experiments, mAP@0.5 and mAP have improved 2.3% and 2.6% respectively, indicating that this modification effectively enhances the feature extraction capability for small objects. At the same time, the FPS has decreased to 351.92 FPS, and GFLOPS has increased, but this does not affect its requirements for industrial applications.

TABLE 6. Ablation study results comparison.

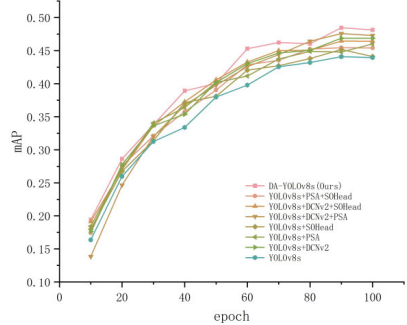
DCNv2	PSA	SOHead	mAP@0.5	mAP	FPS	GLOPS
-	-	-	0.404	0.448	546.83	28.5
-	✓	-	0.415	0.461	506.22	29.6
✓	-	-	0.427	0.474	351.92	25.8
-	-	✓	0.411	0.457	391.72	37.9
✓	✓	-	0.434	0.478	335.99	26.9
✓	-	✓	0.432	0.476	273.19	35.2
-	✓	✓	0.421	0.459	352.18	39.3
✓	✓	✓	0.446	0.486	257.38	36.6

In the Neck network, a PSA self-attention module is added after each C2f module connected to the detection head, reducing information loss in both channel and spatial dimensions and increasing feature fusion capability. We also conducted ablation experiments, where mAP@0.5 and mAP have improved 1.1% and 1.3% respectively. The decrease in FPS and the increase in GFLOPS do not affect the feasibility of its application.

In the Head, we added a small object detection head 4 × 4, which requires an additional structure consisting of Conv, Concat, C2f, and PSA in the Neck network to connect to this detection head. The experiment shows that mAP@0.5 and mAP have improved 0.7% and 0.9% respectively. The FPS still reached 391.72, which does not affect the feasibility of its application.



(a) Comparison of mAP@0.5 across different epochs



(b) Comparison of average precision mAP across different epochs

FIGURE 14. Comparison chart of training accuracy across different epochs.

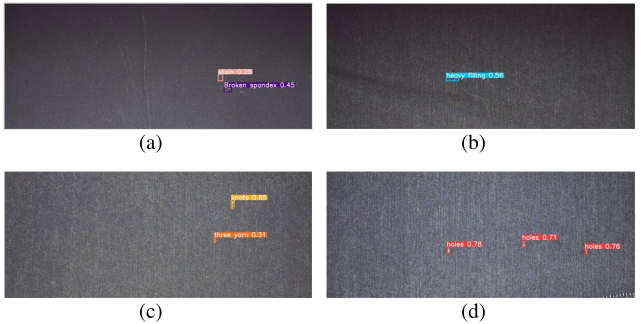


FIGURE 15. Detection results of the DA-YOLOv8s model on the test dataset.

Subsequently, we conducted pairwise combination experiments of DCNv2, PSA, and the small object detection head SOHead on the YOLOv8s benchmark model, which all showed improvements over individual experiments. After applying all three to the model and undergoing 100 epochs of iterative training, mAP@0.5 and mAP were improved by 4.2% and 3.8% respectively, as shown in Table 6 and Fig 14. However, the detection speed FPS evaluation metric decreased by 52.2%, and GFLOPS also increased. But given YOLO's excellent performance in detection speed, DA-YOLOv8s can still achieve 257.38 FPS, which meets the requirements of industrial applications. The partial detection results of the final DA-YOLOv8s on the validation set are shown in Fig 14.

V. CONCLUSION AND FUTURE PROSPECTS

A textile defect detection algorithm based on an improved YOLOv8 algorithm is proposed to address the issues of low detection accuracy and poor real-time performance of traditional methods. Experimental results show that improving the C2f to DCNv2 in the YOLOv8s baseline network can enhance the feature extraction capability of the network, incorporating the self-attention mechanism PSA can increase the feature fusion capability on the channel and spatial dimensions, and adding detection heads can improve the detection ability for small targets. Through these innovations, compared to the basic model, DA-YOLOv8s has improved by 4.2% and 3.8% respectively in mAP@0.5 and mAP. The textile defect detection algorithm designed in this paper has high real-time performance and accuracy, meeting the practical application scenarios of enterprises. The detection results are presented in Fig 15.

Textile defect detection is mostly dominated by small objects, with the detection ability for small targets as a research focus for the next step.

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